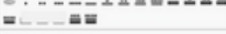


Numeral Systems

What is a Numeral System?

A numeral system is a writing notation to express numbers.

Name	Base	Sample	Approx. first appearance
Babylonian numerals	60		3100 BC
Egyptian numerals	10	 or 	3000 BC
Maya numerals	20		
Indian Numerals	10	० १ २ ३ ४ ५ ६ ७ ८ ९	750 BC – 690 BC
Chinese numerals, Japanese numerals, Korean numerals (Sino-Korean)	10	零 一 二 三 四 五 六 七 八 九	
Roman numerals	10	I II III IV V VI VII VIII IX X	1000 BC
Greek numerals	10	α β γ δ ε ς ζ η θ ι	After 100 BC
Chinese rod numerals	10		1st century
Arabic Numerals	10	0 1 2 3 4 5 6 7 8 9	9th century

Why is base 10 so popular?

Hindu-Arabic Numeral System

Around A.D. 825, the persian mathematician Muḥammad ibn Mūsā al-Khwārizmī wrote a book on algebra that introduced one of the most important inventions in the history of numbers and mathematics.



0

Zero Innovations

Two major consequences derived from zero coming into play that changed the way we deal with numbers forever.

1. For the first time ever there's no special symbol for number ten
2. We now have a positional number system: a particular digit represents a different quantity depending on where it is found in the number

That's how we know that 10 represents a much smaller quantity than 1.000.000

or what allows us to differentiate between 93, 309, and 930

Count me in



The system devised by al-Khwārizmī made it really easy to write a count of any size

Use all the symbols at your disposal until you run out of them

0, 1, 2, 3, 4, 5, 6, 7, 8, 9... now what?

Just add the next available symbol to the left, and repeat the process

10, 11, 12, 13, 14, 15, 16, 17, 18, 19... now what?

20, 21, 22... ok, you've got the general idea

This very simple rule will allow you to count up to any number you can conceive.

Do you read me?



The whole structure of Hindu-Arabic numbers is revealed in the way we pronounce them. Take 4825, for instance. We say "four thousand, eight hundred, twenty-five." That means

four thousands
eight hundreds
two tens and
five.

Or

$$4000 + 800 + 20 + 5$$

Or, even better

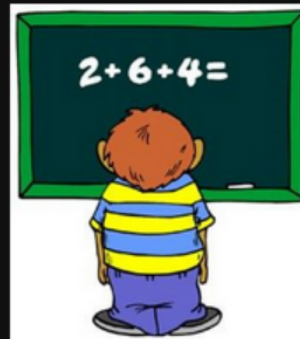
$$\begin{aligned} &4 \times 1000 + \\ &8 \times 100 + \\ &2 \times 10 + \\ &5 \times 1 \end{aligned}$$

It all adds up to...

It's as easy for us to add 3 and 4 as it is to add 30 and 40, or 3000 and 4000.

And it's also not that harder to multiply 40.000 by 20.000 than it is 4 by 2, is it?

Why is that?



It all adds up to...

First of all, because we're dealing with a positional number system

But mostly because some decades ago, you've memorised these two tables

+	1	2	3	4	5	6	7	8	9
1	2	3	4	5	6	7	8	9	10
2	3	4	5	6	7	8	9	10	11
3	4	5	6	7	8	9	10	11	12
4	5	6	7	8	9	10	11	12	13
5	6	7	8	9	10	11	12	13	14
6	7	8	9	10	11	12	13	14	15
7	8	9	10	11	12	13	14	15	16
8	9	10	11	12	13	14	15	16	17
9	10	11	12	13	14	15	16	17	18

Knowing how to add any two digits is what allows you to add two numbers of any size.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	10	12	14	16	18
3	0	3	6	9	12	15	18	21	24	27
4	0	4	8	12	16	20	24	28	32	36
5	0	5	10	15	20	25	30	35	40	45
6	0	6	12	18	24	30	36	42	48	54
7	0	7	14	21	28	35	42	49	56	63
8	0	8	16	24	32	40	48	56	64	72
9	0	9	18	27	36	45	54	63	72	81

There's more...

One last and decisive thing that adds up to the beauty of the positional system is that

Its rules apply as nicely for our counting system as for the ones NOT based on ten

There's nothing magic about number ten (really, there isn't)



Consider the Lobster



What would you say the magic number for lobsters is?

How many different digits do lobsters have to deal with in order to count?

What does 10 mean to a lobster?

The rules for counting, reading, adding you've been using all your life, still apply for lobsters.

{ Board Exercise }

Write from zero to sixteen in base four

{ Board Exercise }

Base four addition table

Now Seriously...

Why do we care?

How many fingers does a computer have?



{ Board Exercise }

Binary to Decimal conversion table

{ Board Exercise }

Number Decomposition

{ Board Exercise }

Powers of Two

Numeral Systems

Computers use binary, can we use something simpler?

Hexadecimal refers to the base-16 number system, which consists of 16 unique symbols

Numbering Systems		
System	Base	Digits
Binary	2	0 1
Octal	8	0 1 2 3 4 5 6 7
Decimal	10	0 1 2 3 4 5 6 7 8 9
Hexadecimal	16	0 1 2 3 4 5 6 7 8 9 A B C D E F

- Numbers 0 to 9 and letters A to F
- Every byte can be represented as two consecutive hexadecimal digits
- Numbers have either an 0x prefix or an h suffix. For example, 0x3F7A
- Much easier to reason about than binary!

{ Board Exercise }

Binary/Decimal/Hexadecimal Conversion Table

{ Board Exercise }

Hex - Binary Conversions

{ Exercise }

Experiment with Desktop Programmer Calculator